

AMENDMENT NO. 1 MARCH 1982

TO

IS : 9456-1980 CODE OF PRACTICE FOR DESIGN AND CONSTRUCTION OF CONICAL AND HYPERBOLIC PARABOLOIDAL TYPES OF SHELL FOUNDATIONS

Alterations

(Page 5, clause 3.1.1, line 1) — Substitute ' (see Fig. 1 and 3) ' for
' (see Fig. 1 to 3) '.

(Page 7, Fig. 3) — Substitute the following for the existing figure:

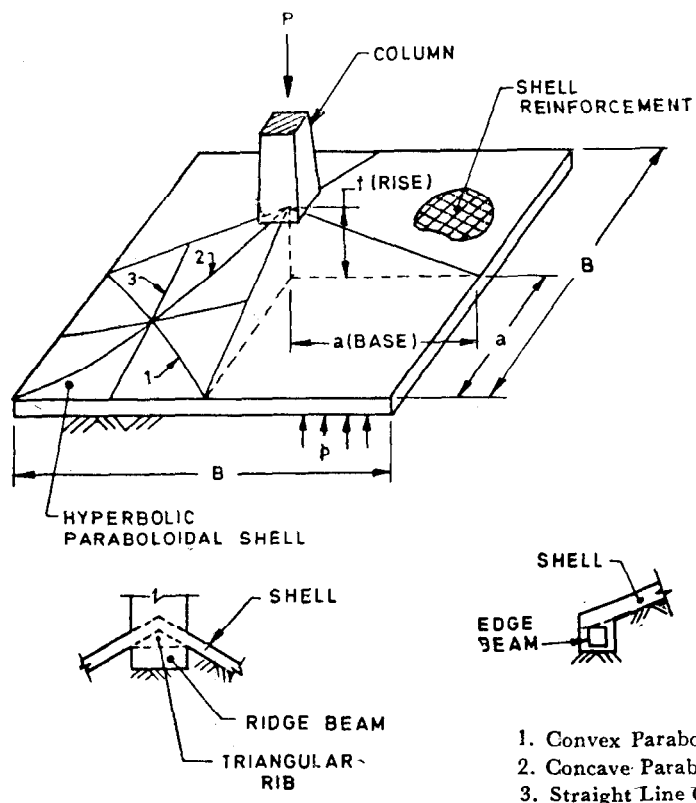


FIG. 3 HYPERBOLIC PARABOLOIDAL SHELL FOOTING

(Page 9, clause 5.4, line 1) — Substitute ' cone and the edge and ' for ' cone edge and the '.

(Page 9, clause 5.4.5, line 2) — Delete the word ' to ' appearing before the word ' delay '.

(Page 11, clause 5.9.2, line 2) — Substitute ' frustrum ' for ' frustrating '.

(Page 11, clause 5.10, line 4) — Substitute ' shear of constant magnitude ' for ' shear '.

(Page 13, clause 5.11, line 9) — Substitute ' so ' for ' to '.

(Page 14, clause 5.12, line 13) — Substitute ' anti-symmetrical ' for ' asymmetrical '.

(Page 19, clause A-1.1.1, value for ' N_θ ') — Substitute the following or the existing value:

$$N_\theta = p_v s \frac{\sin^3 \alpha}{\cos \alpha},$$

(Page 20, Fig. 17) — Substitute the following for the existing figure:

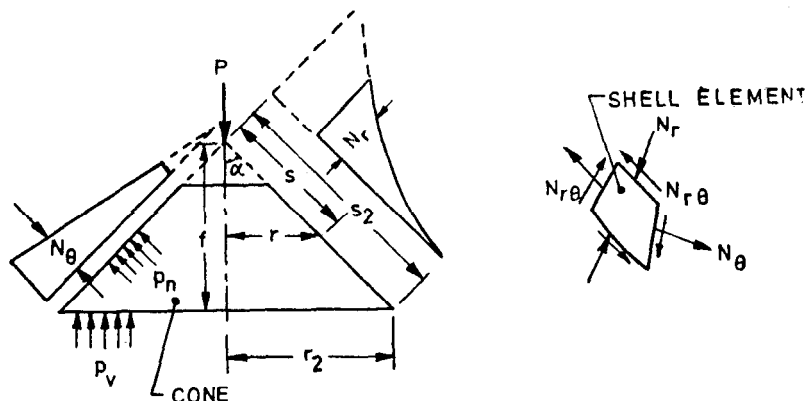


FIG. 17 MEMBRANE STRESS RESULTANTS IN CONICAL FOOTING

(Page 20, clause A-1.1.2, value of ' N_r ') — Substitute the following for the existing value:

$$N_r = \frac{p'_n}{2s} \tan \alpha [s_2^2 - s^2]$$

(Page 21, clause A-1.1.3):

a) Line 1 — Substitute ' Anti-symmetrical ' for ' Asymmetrical '.

b) Value of ' N_θ ' — Substitute the following for the existing value:

$$N_\theta = \frac{p'_n}{s_2} s^2 \tan \alpha \cos \theta$$

- c) *Last line* — Substitute 'anti-symmetrical' for 'asymmetrical'.
- (Page 21, Fig. 19) — Substitute the following for the existing figure:

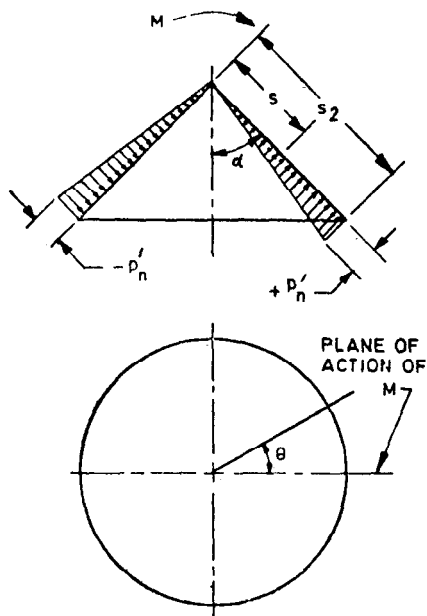


FIG. 19 CONICAL FOOTING UNDER MOMENT

- (Page 22, Fig. 20) — Substitute the following for the existing figure:

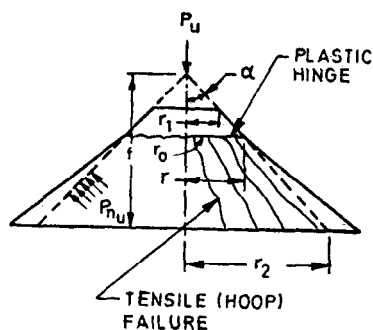


FIG. 20 ULTIMATE FAILURE OF CONICAL FOOTING

(Page 22, clause A-1.2.1, value of ' R_o ') — Substitute the following for the existing value:

' $R_o = \frac{r_o}{r_2}$, where r_o is the radius corresponding to the location of the plastic hinge.'

(Page 22, clause A-1.2.2, value of ' P_u ') — Substitute the following for the existing value:

$$' P_u = p_{n_u} \times A_p '$$

(Page 23, Fig. 21, caption) — Substitute the following for the existing caption:

' FIG. 21 MEMBRANE STRESSES IN HYPAR FOOTING '

(Page 23, clause A-2.1.1, value of ' C ') — Substitute the following for the existing value:

$$' C = 2t \sqrt{a^2 + f^2} '$$

(Page 24, clause A-2.1.2, values for ' N_x ' and ' N_y ') — Substitute the following for the existing values:

$$' N_x = 2 p_n z \sqrt{\frac{1 + k^2 y^2}{1 + k^2 x^2}}$$

$$N_y = 2 p_n z \sqrt{\frac{1 + k^2 x^2}{1 + k^2 y^2}}$$

where

$z = k.x y$, is the co-ordinate of the point (see Fig. 21).

[N_x and N_y are tensile]

$$N_{xy} = \frac{p_n}{2k} \cdot (1 + k^2 x^2 + k^2 y^2)$$

(Page 25, clause A-2.2.1, value of ' P_u ') — Substitute the following for the existing value:

$$' P_u = 12 N_f \left[\left\{ \frac{a}{f} + \frac{1}{2} \left(\frac{a}{f} \right)^3 \right\} \left\{ \log_e \left(\frac{f}{a} \right) + \sqrt{1 + \left(\frac{f}{a} \right)^2} \right\} \right. \\ \left. - \frac{1}{2} \left(\frac{a}{f} \right)^3 \left\{ \frac{f}{a} \times \sqrt{1 + \left(\frac{f}{a} \right)^2} \right\} \right] + 12 N_b \frac{f}{a} + 6 \frac{M_r}{a} '$$

(Page 26, clause A-2.2.2, value of ' P_u ') — Substitute the following for the existing value:

$$' P_u = 4 N_f + 8 N_b \left(\frac{f}{a} \right) + \frac{8}{\sqrt{2}} \frac{M'_r}{a} '$$

(*Page 28, Appendix C, line 5*) — Substitute ‘ air-borne ’ for
‘ airborne ’.

Addendum

(*Page 4, clause 0.4*) — Add the following new note after 0.4:

‘NOTE — The provisions given in this standard have been explained in detail
in the book ‘Modern Foundation — An Introduction to Advanced Techni-
ques: Part I Shell Foundation’ by Dr Nainan P. Kurian.’

(BDC 43)